1.7 Differentiation

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

Differentiation is the branch of calculus that measures how quickly one quantity changes in response to changes in another. Geometrically, the derivative of a function at a point is the gradient of the tangent line to its graph there; dynamically, it is the instantaneous rate of change. This chapter develops the machinery for computing derivatives — first principles, the standard-functions table, the product, quotient and chain rules, implicit and parametric differentiation — and uses them to classify stationary points, write down tangents and normals, and solve related-rates problems. The derivative is the single most useful tool in applied mathematics: every optimisation problem, every motion problem, and every model involving change ultimately rests on it.

1. The Gradient at a Point

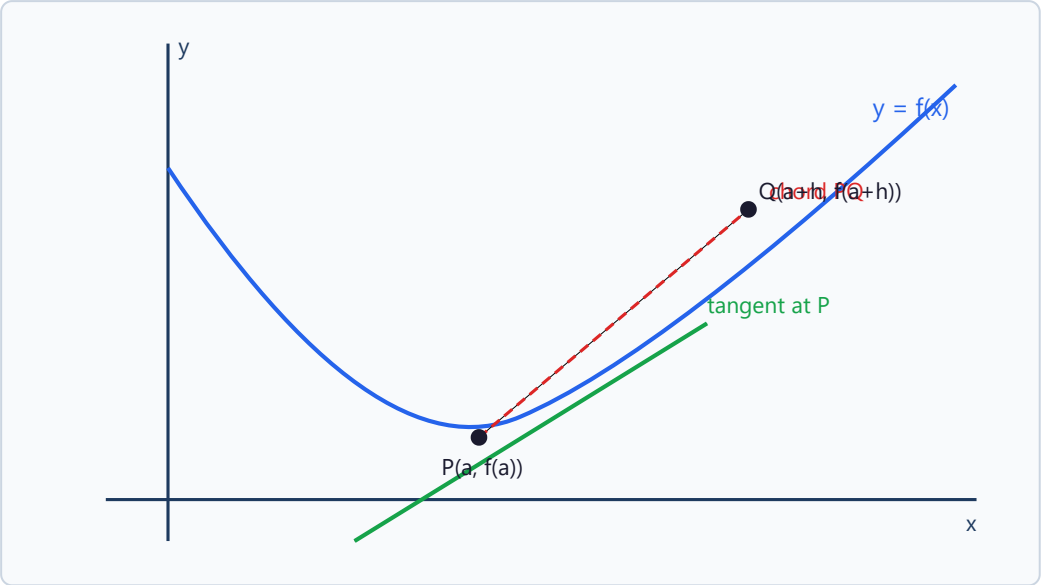
For a straight line the gradient is constant and easy to measure: rise over run between any two points. For a curve the gradient varies from point to point, so "the gradient of the curve" has to be defined *at* a specific point. The key idea is that, on a sufficiently small scale, any smooth curve looks locally like a straight line — the tangent line. The gradient of that tangent is what we call the gradient of the curve at the point.

> **Tangent and gradient of a curve.** The *tangent* to a curve $y = f(x)$ at the point $P(a, f(a))$ is the straight line that touches the curve at P and has the same direction as the curve there. The *gradient of the curve at* P is the gradient of this tangent. Intuitively, it is the instantaneous rate at which y changes with x at that point.

From Average Rate to Instantaneous Rate

Take two points on the curve, $P(a, f(a))$ and $Q(a + h, f(a + h))$. The chord PQ has gradient \[\frac{f(a + h) - f(a)}{h}, \] the *average rate of change* of f over the interval $[a, a + h]$. As Q is brought

closer to P (i.e. $h \to 0$), the chord rotates about P and approaches the tangent. The limiting value of the chord gradient, if it exists, is the gradient of the tangent — the instantaneous rate of change at P.



This limiting process — *differentiation from first principles* — is the rigorous foundation on which every later shortcut (power rule, product rule, etc.) is built.

2. Differentiation from First Principles

Derivative (first principles). The *derivative* of f at x, written $f'(x)$ or $\dfrac{dy}{dx}$, is the limit \[f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}, \] provided this limit exists. If it does, f is said to be *differentiable* at x. The function f' that sends each x to this value is the *derivative function*.

Three equivalent notations are used interchangeably: $f'(x)$ (Lagrange), $\dfrac{dy}{dx}$ (Leibniz), and $\dot{y}$ (Newton, usually reserved for derivatives with respect to time). Each has its place: Leibniz is best for the chain rule; Lagrange is concise; Newton is standard in mechanics.

Worked Example 1 — Derive d/dx(x²) = 2x from first principles

```
Let f(x) = x². Apply the definition:
   f'(x) = lim_(h→0) [f(x + h) − f(x)] / h
         = lim_(h→0) [(x + h)² − x²] / h.
Expand the numerator:
```

```
5.     (x + h)² − x² = x² + 2xh + h² − x² = 2xh + h².
6.  Divide by h (valid while h ≠ 0):
7.     (2xh + h²) / h = 2x + h.
8.  Take the limit h → 0:
9.     lim_(h→0) (2x + h) = 2x.
10. ∴ d/dx(x²) = 2x.

// The same procedure gives d/dx(x³) = 3x² and, by induction, d/dx(xⁿ) = n x^(n−1
// for every positive integer n. The binomial theorem generalises it to all real
```

Worked Example 2 — General power rule for small n

Show $d/dx(x^3) = 3x^2$ from first principles.

```
1.  f(x) = x³, so f(x + h) = (x + h)³.
2.  Expand by the binomial theorem:
3.     (x + h)³ = x³ + 3x²h + 3xh² + h³.
4.  Subtract f(x) and divide by h:
5.     [f(x + h) − f(x)] / h = (3x²h + 3xh² + h³) / h
6.                            = 3x² + 3xh + h².
7.  Let h → 0:  every term with an h vanishes.
8.     f'(x) = 3x².

// The general formula d/dx(xⁿ) = n x^(n−1) follows because in (x + h)ⁿ
// the first two binomial terms are xⁿ + n x^(n−1) h; the rest all carry h².
```

Exam tip — first-principles questions are syllabus-required. OCR routinely asks you to "differentiate from first principles", usually for a polynomial of degree ≤ 3. Always (i) write the limit definition explicitly, (ii) expand the numerator, (iii) cancel the h in the denominator *before* taking the limit, and (iv) state "as $h \to 0$" at the final step. Skipping the limit statement loses a mark even if the algebra is correct.

3. Basic Rules of Differentiation

Three linearity rules let you differentiate any polynomial or sum of power functions term-by-term.

Power rule. For any real n, \[\frac{d}{dx}(x^n) = n x^{n-1}. \] The rule holds for positive integers (proved above), negative integers (so $\frac{d}{dx}(1/x) = -1/x^2$), fractional powers (so $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2}x^{-1/2}$), and even irrational powers.

Sum, difference and constant-multiple rules. For differentiable f, g and constant c: \[(f + g)'(x) = f'(x) + g'(x), \quad (f - g)'(x) = f'(x) - g'(x), \quad (cf)'(x) = c f'(x). \] In particular, $\dfrac{d}{dx}(c) = 0$ for any constant c.

Rule	Statement	Example
Power	$(x^n)' = nx^{n-1}$	$(x^5)' = 5x^4$; $(x^{-2})' = -2x^{-3}$
Constant multiple	$(cf)' = cf'$	$(7x^3)' = 21x^2$
Sum / difference	$(f \pm g)' = f' \pm g'$	$(x^3 - 4x)' = 3x^2 - 4$
Constant	$(c)' = 0$	$(\pi)' = 0$
Linear	$(ax + b)' = a$	$(5x - 3)' = 5$

Worked Example 3 — Polynomial with mixed powers

Differentiate $y = 4x^3 - 6\sqrt{x} + \dfrac{2}{x^2} + 7$.

```
1.  Rewrite everything as a power:
2.     y = 4x³ − 6 x^(1/2) + 2 x^(−2) + 7.
3.  Differentiate term by term using power + constant-multiple rules:
4.     dy/dx = 12 x² − 3 x^(−1/2) − 4 x^(−3) + 0.
5.  Rewrite in original style if required:
6.     dy/dx = 12x² − 3/√x − 4/x³.
```

```
// Always convert roots and fractions to index notation BEFORE applying power rul
```

Exam tip — rewrite before differentiating. Many wrong answers arise because a student tried to differentiate $1/x^3$ or $\sqrt{x}$ directly. Always rewrite as x^{-3} or $x^{1/2}$ first — the power rule handles these effortlessly, while the original notation invites careless mistakes. The same applies to products inside fractions: simplify algebraically before reaching for the quotient rule if you can.

4. Derivatives of Standard Functions

Beyond polynomials, a handful of standard functions — trigonometric, exponential, logarithmic — have known derivatives that must be memorised. Each can be derived from first principles (some use the addition formulas; others require limits like $\lim_{h\to 0} \sin h/h = 1$), but in exams they are quoted directly.

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\tan x) = \sec^2 x.$$

$$\frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(\ln x) = \frac{1}{x}, \quad \frac{d}{dx}(a^x) = a^x \ln a.$$

A mnemonic for the trigonometric pair: differentiating $\sin$ and $\cos$ cycles through the pattern $\sin \to \cos \to -\sin \to -\cos \to \sin$. The minus sign appears only when passing through $\cos$.

Function $f(x)$	Derivative $f'(x)$	Notes
x^n	nx^{n-1}	Any real n.
$\sin x$	$\cos x$	x must be in radians.
$\cos x$	$-\sin x$	Minus sign — easily forgotten.
$\tan x$	$\sec^2 x = 1/\cos^2 x$	Undefined where $\cos x = 0$.
e^x	e^x	Unique: derivative equals the function.

$\ln x$	$1/x$	Domain $x > 0$.
a^x	$a^x \ln a$	Extra factor of $\ln a$ for $a \neq e$.
$\log_a x$	$1/(x \ln a)$	Becomes $1/x$ when $a = e$.

Exam tip — trig derivatives require radians. Every derivative formula for $\sin, \cos, \tan$ assumes x is measured in radians. If you differentiate $\sin x$ thinking in degrees, the answer is wrong by a factor of $\pi/180$. OCR papers always work in radians unless explicitly stated — set your calculator to RAD mode before starting any calculus question.

5. The Product Rule

When y is the product of two functions of x, its derivative is *not* the product of the individual derivatives. Instead:

Product rule. If u, v are differentiable functions of x, then \[\frac{d}{dx}(u v) = u' v + u v' \quad\text{or, in Leibniz form,}\quad \frac{d}{dx}(uv) = v\frac{du}{dx} + u\frac{dv}{dx}. \] Mnemonic: "derivative of the first times the second, plus the first times the derivative of the second".

The rule extends by induction to products of three or more factors: $(uvw)' = u'vw + uv'w + uvw'$ — each term differentiates one factor while leaving the others alone.

Worked Example 4 — d/dx(x · sin x)

```
1.  Identify u = x, v = sin x.
2.  Compute the pieces:
3.     u' = 1        v' = cos x.
4.  Apply the product rule:
5.     (uv)' = u'v + uv'
6.           = (1)(sin x) + (x)(cos x)
7.           = sin x + x cos x.
```

Worked Example 5 — d/dx(e^x · cos x)

```
1.  u = eˣ, v = cos x.
2.  u' = eˣ;  v' = −sin x.
3.  (uv)' = eˣ · cos x + eˣ · (−sin x)
4.        = eˣ (cos x − sin x).

// Factoring out eˣ gives a clean, simplified answer — examiners prefer this.
```

Worked Example 6 — d/dx((x² + 1)(x³ − 2))

```
1.  u = x² + 1,  v = x³ − 2.
2.  u' = 2x,       v' = 3x².
3.  (uv)' = (2x)(x³ − 2) + (x² + 1)(3x²)
4.        = 2x⁴ − 4x + 3x⁴ + 3x²
5.        = 5x⁴ + 3x² − 4x.
6.  Check by expanding first:  (x² + 1)(x³ − 2) = x⁵ − 2x² + x³ − 2.
7.         Differentiate: 5x⁴ − 4x + 3x² ✓  (same answer.)

// For polynomial products, expanding first is often quicker; the product rule
// is essential only when expansion is impossible (e.g. x · sin x).
```

6. The Quotient Rule

Quotient rule. If u, v are differentiable and $v \neq 0$, then \[\frac{d}{dx}\!\left(\frac{u}{v}\right) = \frac{u' v - u v'}{v^2}. \] Mnemonic: "bottom times derivative-of-top, minus top times derivative-of-bottom, all over bottom squared." The order $u'v - uv'$ matters — swapping the subtraction sign gives the wrong answer.

The quotient rule can always be derived from the product rule by writing $u/v = u \cdot v^{-1}$ and applying the product and chain rules — but memorising it directly is faster in exams.

Worked Example 7 — d/dx(sin x / x)

```
1.  u = sin x,  v = x.
2.  u' = cos x,  v' = 1.
3.  (u/v)' = (u'v − uv') / v²
4.         = (cos x · x − sin x · 1) / x²
5.         = (x cos x − sin x) / x².

// This derivative appears in the analysis of the "sinc" function used in signal
// processing. As x → 0, (sin x)/x → 1, but its derivative at 0 is zero.
```

Worked Example 8 — d/dx((x² + 1)/(x − 1))

```
1.  u = x² + 1,  v = x − 1.
2.  u' = 2x,       v' = 1.
3.  (u/v)' = [2x(x − 1) − (x² + 1)(1)] / (x − 1)²
4.         = [2x² − 2x − x² − 1] / (x − 1)²
5.         = (x² − 2x − 1) / (x − 1)².
6.  Domain: x ≠ 1 (where the original quotient is undefined).
```

Exam tip — don't mix up product and quotient order. Both rules contain "$u'v$" and "uv'", but the product rule adds, while the quotient rule subtracts — and the quotient rule's order is $u'v - uv'$, not $uv' - u'v$. A useful check: on the quotient rule, the term with the derivative of the *numerator* comes first (positive); the term with the derivative of the *denominator* carries the minus sign.

7. The Chain Rule

Most functions met in practice are composites: $\sin(3x)$, e^{x^2}, $(2x+1)^5$, $\ln(x^2+1)$. To differentiate a composite $y = f(g(x))$ you need the chain rule.

Chain rule. If $y = f(u)$ and $u = g(x)$, both differentiable, then \[\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}, \qquad\text{equivalently}\qquad \big(f(g(x))\big)' = f'(g(x)) \cdot g'(x). \] Intuition: differentiate the *outer* function (treating the inside as a single blob), then multiply by the derivative of the inside.

In Leibniz notation the rule looks like cancellation of du: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$. This is a deliberate feature of Leibniz's notation — it behaves like a fraction for this purpose, even though $\frac{dy}{dx}$ is not literally a ratio.

Worked Example 9 — Four chain-rule calculations

```
1.  d/dx (sin 3x):  outer = sin, inner = 3x.
2.     Derivative = cos(3x) · 3 = 3 cos 3x.
3.
4.  d/dx (e^(x²)):  outer = eᵘ, inner = x².
5.     Derivative = e^(x²) · 2x = 2x e^(x²).
6.
7.  d/dx ((2x + 1)⁵):  outer = u⁵, inner = 2x + 1.
8.     Derivative = 5(2x + 1)⁴ · 2 = 10(2x + 1)⁴.
9.
10. d/dx (ln(x² + 1)):  outer = ln u, inner = x² + 1.
11.    Derivative = [1/(x² + 1)] · 2x = 2x / (x² + 1).

// Pattern: "derivative of outer at the inside, times derivative of inside".
// When the inner is linear (ax + b), the chain rule adds only a factor of a.
```

Exam tip — the chain-rule multiplier is easy to miss. Students who see $y = \sin(3x)$ often write $dy/dx = \cos(3x)$ and lose the factor of 3. A quick check: when the inner function is a linear term $ax + b$, the chain rule contributes the factor a. So $(e^{5x})' = 5e^{5x}$; $(\cos(2x - 1))' = -2\sin(2x - 1)$; $(\ln(4x))' = 4/(4x) = 1/x$ — the factor sometimes cancels, but you still had to include it.

Exam tip — chain rule sign traps. The derivative of $\cos(\text{stuff})$ is $-\sin(\text{stuff}) \cdot (\text{stuff})'$ — the minus sign comes from differentiating the outer cosine, not from the inner function. Even when the

inner function is $-x$, the minus signs multiply: $(\cos(-x))' = -\sin(-x) \cdot (-1) = \sin(-x) \cdot 1 = -\sin x$ (using $\sin(-x) = -\sin x$). Work systematically with brackets when signs get confusing.

8. Implicit Differentiation

Sometimes y is defined implicitly by an equation in x and y, without an explicit formula $y = f(x)$. Examples include the circle $x^2 + y^2 = 25$ and the folium $x^3 + y^3 = 3xy$. To find dy/dx, differentiate both sides with respect to x, treating y as an unknown function of x and applying the chain rule to every y -term.

Implicit differentiation. Given a relation $F(x, y) = 0$, differentiate each term with respect to x; wherever a y appears, multiply by $\dfrac{dy}{dx}$ (chain rule). Then solve algebraically for $\dfrac{dy}{dx}$. Key building blocks: \[\frac{d}{dx}(y^n) = n y^{n-1} \frac{dy}{dx}, \qquad \frac{d}{dx}(xy) = y + x\frac{dy}{dx}, \qquad \frac{d}{dx}(\sin y) = \cos y \cdot \frac{dy}{dx}. \]

Worked Example 10 — Circle $x^2 + y^2 = 25$

```
1.  Differentiate each term w.r.t. x:
2.     d/dx(x²) + d/dx(y²) = d/dx(25)
3.     2x + 2y · dy/dx = 0.
4.  Solve for dy/dx:
5.     2y · dy/dx = -2x
6.     dy/dx = -x/y      (provided y ≠ 0).
7.  Check at (3, 4):  dy/dx = -3/4.
8.  Geometric check: radius OP has gradient 4/3; tangent ⊥ radius, so tangent
9.  gradient = -3/4 ✓.

// The tangent to a circle is perpendicular to the radius — this implicit result
// confirms it without any Pythagorean manipulation.
```

Worked Example 11 — Implicit with a product term

Given $x^2y + xy^2 = 6$, find $\frac{dy}{dx}$.

```
1.  Differentiate each term w.r.t. x, using the product rule on xⁿyᵐ terms.
2.  Term 1: d/dx(x²y) = 2x · y + x² · dy/dx.
3.  Term 2: d/dx(xy²) = y² + x · 2y · dy/dx = y² + 2xy · dy/dx.
4.  RHS:    d/dx(6) = 0.
5.  Combine:
6.     2xy + x² dy/dx + y² + 2xy · dy/dx = 0.
7.  Collect dy/dx terms:
8.     (x² + 2xy) dy/dx = -(2xy + y²)
9.     dy/dx = -(2xy + y²) / (x² + 2xy)
10.          = -y(2x + y) / [x(x + 2y)].

// Final answer simplified by factoring numerator and denominator.
```

Exam tip — never forget the dy/dx factor. Every time you differentiate a term containing y, the chain rule contributes a factor of dy/dx. A missed dy/dx is the single commonest error in implicit-differentiation questions. After differentiating, every term that originally contained y should now contain either dy/dx or an explicit y — if neither appears, you have made an algebra slip.

9. Parametric Differentiation

A curve may be given parametrically as $x = f(t)$, $y = g(t)$, with t the parameter. To find dy/dx in terms of t, differentiate both coordinates with respect to t and divide.

Parametric differentiation formula. For a curve $x = f(t)$, $y = g(t)$ with $f'(t) \neq 0$, \[\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}. \] The formula follows from the chain rule: $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$, so rearranging gives the quoted identity.

Worked Example 12 — Circle x = r cos t, y = r sin t

```
dx/dt = −r sin t;     dy/dt = r cos t.
dy/dx = (dy/dt) / (dx/dt)
        = (r cos t) / (−r sin t)
        = −cos t / sin t
        = −cot t.
Sanity check at t = π/2 (top of circle, point (0, r)):
   dy/dx = −cot(π/2) = 0        ✓  (horizontal tangent.)
At t = π (leftmost point (−r, 0)):  dy/dx undefined (dx/dt = 0,
vertical tangent.)
```

Worked Example 13 — x = t² + 1, y = 2t

```
dx/dt = 2t;   dy/dt = 2.
dy/dx = 2 / (2t) = 1/t     (valid for t ≠ 0).
Eliminate parameter to check:
   t = y/2  ⇒  x = (y/2)² + 1 = y²/4 + 1,
   ⇒  y² = 4(x − 1)      (a parabola opening right.)
Differentiate implicitly: 2y · dy/dx = 4 ⇒ dy/dx = 2/y = 2/(2t) = 1/t ✓.

// Two routes (parametric and implicit) give the same answer, as they must.
```

10. The Second Derivative

Differentiating $f'(x)$ gives the *second derivative* $f''(x) = \dfrac{d^2y}{dx^2}$. Geometrically, it measures how the gradient is itself changing: the *rate of change of the rate of change*. In mechanics, if $s(t)$ is displacement, $s'(t)$ is velocity and $s''(t)$ is acceleration.

Second derivative and concavity. The second derivative $f''(x)$ describes the *curvature* of the graph:

- $f''(x) > 0$: graph is *concave up* (shaped like ∪) — tangent lines lie below the curve.
- $f''(x) < 0$: graph is *concave down* (shaped like ∩) — tangent lines lie above the curve.
- $f''(x) = 0$ with sign change: a *point of inflection*.

Example: for $y = x^3$, $y' = 3x^2$, $y'' = 6x$. At $x = 0$, $y'' = 0$ and changes from negative (for $x < 0$) to positive (for $x > 0$) — the origin is a point of inflection.

11. Stationary Points

A point on the curve where the tangent is horizontal is called a *stationary point*. These are the candidates for local maxima, local minima, and certain kinds of inflection.

Stationary point. A stationary point of f is a value x_0 where $f'(x_0) = 0$. The point on the graph is $(x_0, f(x_0))$. Stationary points fall into three categories:

- **Local maximum:** f' changes from positive to negative at x_0.
- **Local minimum:** f' changes from negative to positive at x_0.
- **Stationary point of inflection:** f' does *not* change sign at x_0 (e.g. $y = x^3$ at origin).

Classifying Stationary Points

Two tests are standard: the first-derivative (sign-change) test and the second-derivative test.

Test	How it works	Conclusion
First-derivative (sign change)	Examine sign of f' just before and just after x_0.	$+$ then $-$: max. $-$ then $+$: min. Same sign either side: inflection.
Second-derivative	Compute $f''(x_0)$.	$f''(x_0) > 0$: min. $f''(x_0) < 0$: max. $f''(x_0) = 0$: inconclusive.

Exam tip — when $f''(x_0) = 0$, the second-derivative test FAILS. It does not mean "inflection" — it means "try something else". For $y = x^4$ at $x = 0$: $f'(0) = 0$ and $f''(0) = 0$, but the origin is a

minimum (the graph sits on the x-axis and rises on both sides). Switch to the first-derivative test: $f'(x) = 4x^3$ is negative for $x < 0$ and positive for $x > 0$, confirming a minimum.

12. Increasing and Decreasing Functions

Monotonicity from the derivative. On an interval I:

- If $f'(x) > 0$ for all $x \in I$, then f is *strictly increasing* on I.
- If $f'(x) < 0$ for all $x \in I$, then f is *strictly decreasing* on I.
- If $f'(x) \geq 0$ throughout I (equality only at isolated points), then f is (non-strictly) increasing.

To find the intervals on which a function is increasing or decreasing, solve the inequalities $f'(x) > 0$ and $f'(x) < 0$ — boundary points are the stationary points.

13. Points of Inflection

A *point of inflection* is where the concavity of the graph changes — i.e. where the second derivative changes sign. It need not be a stationary point (e.g. $y = x^3 + x$ has an inflection at $x = 0$ but $y'(0) = 1 \neq 0$).

Point of inflection. A point $(x_0, f(x_0))$ is a point of inflection if $f''(x_0) = 0$ *and* f'' changes sign at x_0. Additional cases:

- **Stationary point of inflection:** $f'(x_0) = 0$ *and* $f''(x_0) = 0$ *and* f'' changes sign. Example: $y = x^3$ at origin.
- **Non-stationary point of inflection:** $f''(x_0) = 0$ with f'' changing sign, but $f'(x_0) \neq 0$.

The condition "f'' changes sign" is essential: $y = x^4$ has $f''(0) = 0$ but no inflection there.

14. Tangents and Normals

The tangent at $(a, f(a))$ passes through that point with gradient $f'(a)$. The normal is the line through the same point perpendicular to the tangent.

$$\text{Tangent:} \quad y - f(a) = f'(a)\,(x - a).$$

$$\text{Normal:} \quad y - f(a) = -\frac{1}{f'(a)}\,(x - a) \quad (f'(a) \neq 0).$$

If $f'(a) = 0$, the tangent is horizontal ($y = f(a)$) and the normal is vertical ($x = a$).

15. Related Rates

If two quantities are both functions of time, the chain rule relates their rates of change. The pattern: differentiate an equation linking them w.r.t. t, substitute known values, solve for the unknown rate.

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

Typical contexts: expanding balloons (volume vs radius), cones filling with water (volume vs height), ladders sliding down walls (positions at the two ends).

16. Applied Worked Examples

Worked Example 14 — Classifying stationary points of $y = x^3 - 3x^2 + 2$

```
Differentiate:
   dy/dx = 3x² − 6x = 3x(x − 2).
Stationary points: 3x(x − 2) = 0  ⇒  x = 0 or x = 2.
y-values:
   x = 0:  y = 0 − 0 + 2 = 2.    Point (0, 2).
   x = 2:  y = 8 − 12 + 2 = −2.  Point (2, −2).
Second derivative: d²y/dx² = 6x − 6.
At x = 0: d²y/dx² = −6 < 0  ⇒  local MAXIMUM at (0, 2).
At x = 2: d²y/dx² = 6 > 0   ⇒  local MINIMUM at (2, −2).
```

```
// Inflection: d²y/dx² = 0 at x = 1. Sign: negative for x<1, positive for x>1,
// so (1, 0) is a (non-stationary) point of inflection.
```

Worked Example 15 — Tangent to y = 2 sin x at x = π/6

```
1.  Find the point:
2.     y(π/6) = 2 sin(π/6) = 2 · ½ = 1.       Point (π/6, 1).
3.  Find the gradient:
4.     dy/dx = 2 cos x.
5.     at x = π/6:  dy/dx = 2 cos(π/6) = 2 · (√3/2) = √3.
6.  Tangent equation:
7.     y − 1 = √3 (x − π/6)
8.     y = √3 · x − (√3 π)/6 + 1.

// Normal (perpendicular): gradient = −1/√3.
// y − 1 = −(1/√3)(x − π/6).
```

Worked Example 16 — Cone filling with water (related rates)

A cone has radius r and height h with $r = h$ (i.e. half-angle 45°). Water is added so that the volume increases at $5\,\mathrm{cm}^3/\mathrm{s}$. How fast is h increasing when $h = 10\,\mathrm{cm}$?

```
1.  Volume of a cone:  V = (1/3) π r² h.
2.  Use the constraint r = h to eliminate r:
3.     V = (1/3) π h² · h = (π/3) h³.
4.  Differentiate w.r.t. time t:
5.     dV/dt = π h² · dh/dt.
6.  Substitute dV/dt = 5, h = 10:
7.     5 = π · 100 · dh/dt
8.     dh/dt = 5 / (100π) = 1 / (20π) cm/s.
9.  Numerical value:
10.    dh/dt ≈ 0.0159 cm/s (≈ 1.59 × 10⁻² cm/s).
```

```
// Units: dV/dt in cm³/s; h² in cm²; so dh/dt in cm/s ✓.
// Physically the water level rises slowly — most of the inflow is spreading outw
```

Worked Example 17 — Triple product: y = x · e^(2x) · ln x

```
1.  Three-factor product rule:
2.     (uvw)' = u'vw + uv'w + uvw'.
3.  Take u = x, v = e^(2x), w = ln x.
4.     u' = 1
5.     v' = 2 e^(2x)         (chain rule)
6.     w' = 1/x.
7.  Assemble:
8.     dy/dx = (1) · e^(2x) · ln x
9.           + x · 2 e^(2x) · ln x
10.          + x · e^(2x) · (1/x)
11.         = e^(2x) ln x + 2x e^(2x) ln x + e^(2x)
12.         = e^(2x) [ ln x + 2x ln x + 1 ]
13.         = e^(2x) [ (1 + 2x) ln x + 1 ].

// Always factor out the common e^(2x) — tidy form is easier to differentiate
// again or to substitute values into.
```

Worked Example 18 — Parametric, second derivative: x = 2t, y = 4t² + 3

```
1.  First derivatives w.r.t. t:
2.     dx/dt = 2;      dy/dt = 8t.
3.  First derivative dy/dx:
4.     dy/dx = (dy/dt) / (dx/dt) = 8t / 2 = 4t.
5.  At t = 1:  dy/dx = 4.
6.
7.  For the second derivative, differentiate dy/dx = 4t w.r.t. x (NOT t):
8.     d²y/dx² = d/dx (4t) = (d/dt (4t)) / (dx/dt) = 4 / 2 = 2.
9.  At t = 1:  d²y/dx² = 2.
10.
```

```
Sanity check: eliminate t. t = x/2, so y = 4(x/2)² + 3 = x² + 3.
Direct: dy/dx = 2x; at x = 2 (t = 1), dy/dx = 4 ✓.
        d²y/dx² = 2 ✓.

// For parametric second derivatives, NEVER just differentiate dy/dx w.r.t. t.
// You must divide by dx/dt afterwards — a common slip that costs marks.
```

17. Consolidated Summary and Exam Strategy

Situation	Technique	Key formula
Polynomial / power function	Power rule, term-by-term	$(x^n)' = nx^{n-1}$
Trig / exp / log	Standard derivatives (quote)	See table in §4.
Product of two functions	Product rule	$(uv)' = u'v + uv'$
Quotient	Quotient rule	$(u/v)' = (u'v - uv')/v^2$
Composite $f(g(x))$	Chain rule	$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$
Implicit relation	Differentiate both sides; chain rule on y	Each y-term picks up dy/dx.
Parametric	Divide: $dy/dx = (dy/dt)/(dx/dt)$	$d^2y/dx^2 = \frac{d}{dt}(dy/dx) \div dx/dt$
Classify stationary point	Second-deriv test, fallback to sign change	$f'' > 0$ min; $f'' < 0$ max.
Related rates	Differentiate constraint w.r.t. t	$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$

Chapter-wide checklist for Differentiation.

- **Know the standard derivatives by heart:** $x^n, \sin, \cos, \tan, e^x, \ln x, a^x$. Quoting them costs no marks; deriving them live costs time.

- **First principles** must show the limit definition, the algebraic simplification, and the statement "as $h \to 0$".
- **Simplify BEFORE differentiating** where possible — rewrite roots and fractions as indices.
- **Chain rule: don't forget the inner factor.** $(\sin 3x)' = 3\cos 3x$, not $\cos 3x$.
- **Implicit: every y-term gains dy/dx.** Then solve algebraically.
- **Parametric second derivative:** differentiate dy/dx w.r.t. t and divide by dx/dt.
- **Classify stationary points** with f'' first; if it's zero, use first-derivative sign test.
- **Tangent:** $y - y_0 = m(x - x_0)$**; normal gradient** $= -1/m$, provided $m \neq 0$.
- **Always use radians** for any calculus involving trig.
- **State units** in related-rates and applied problems.

Typical OCR examiner complaints (from mark reports).

- Skipping the "$\lim_{h\to 0}$" in first-principles answers.
- Forgetting the chain-rule inner derivative: writing $(e^{x^2})' = e^{x^2}$ instead of $2xe^{x^2}$.
- Misapplying the quotient rule order — writing $uv' - u'v$ instead of $u'v - uv'$.
- Losing the dy/dx factor on one or more implicit terms.
- Concluding "inflection" the moment $f''(x_0) = 0$, without checking that f'' changes sign.
- In parametric questions, differentiating dy/dx w.r.t. t and calling that d^2y/dx^2 (wrong — must divide by dx/dt).
- Working in degrees by mistake, giving $(\sin x)' = \cos x \cdot \pi/180$ that goes uncaught.
- Giving a tangent equation at a point where f is not defined, or a normal where $f'(a) = 0$ (normal is vertical, not "undefined gradient" line).

Formula snapshot (memorise).

- First principles: $f'(x) = \lim_{h\to 0}[f(x+h) - f(x)]/h$.
- Power: $(x^n)' = nx^{n-1}$. Sum/difference/constant-multiple: linear.
- Trig: $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$; $(\tan x)' = \sec^2 x$.
- Exp/log: $(e^x)' = e^x$; $(\ln x)' = 1/x$; $(a^x)' = a^x \ln a$.

- Product: $(uv)' = u'v + uv'$.
- Quotient: $(u/v)' = (u'v - uv')/v^2$.
- Chain: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$.
- Implicit: each y-term on differentiation carries a factor dy/dx.
- Parametric: $dy/dx = (dy/dt)/(dx/dt)$; $d^2y/dx^2 = \frac{d}{dt}(dy/dx)/(dx/dt)$.
- Stationary classification: $f'' > 0$ min; $f'' < 0$ max; $f'' = 0$ inconclusive.
- Tangent: $y - f(a) = f'(a)(x - a)$. Normal: $y - f(a) = -(1/f'(a))(x - a)$.